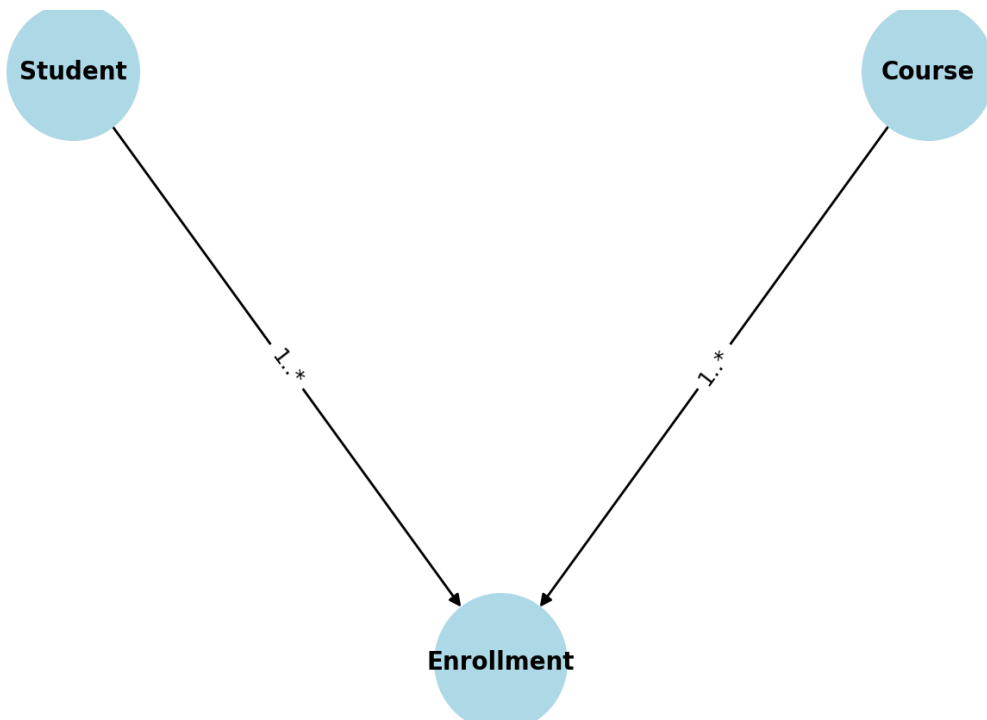


Student Database: Implementation and Manipulation in SQL Server

Entity-Relationship Diagram (ERD)

The following ER diagram represents the Student database system with three entities: Student, Course, and Enrollment.



1. Creating the Database and Tables

```
CREATE DATABASE Student;  
USE Student;
```

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY IDENTITY(1,1),  
    Name NVARCHAR(100),  
    FatherName NVARCHAR(100),  
    FatherOccupation NVARCHAR(100),  
    District NVARCHAR(50),  
    Department NVARCHAR(50),  
    CGPA DECIMAL(3,2)  
);
```

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY IDENTITY(1,1),  
    CourseName NVARCHAR(100),  
    Credits INT  
);
```

```
CREATE TABLE Enrollment (  
    EnrollmentID INT PRIMARY KEY IDENTITY(1,1),  
    StudentID INT FOREIGN KEY REFERENCES Student(StudentID),  
    CourseID INT FOREIGN KEY REFERENCES Course(CourseID),  
    Grade CHAR(2)  
);
```

2. Inserting Data

```
INSERT INTO Student (Name, FatherName, FatherOccupation, District, Department, CGPA)
VALUES
('Rahim Uddin', 'Karim Uddin', 'Teacher', 'Dhaka', 'CSE', 3.75),
('Sadia Akter', 'Shafiq Akter', 'Doctor', 'Chittagong', 'EEE', 3.60),
('Mizan Rahman', 'Jamal Rahman', 'Businessman', 'Dhaka', 'BBA', 3.20);
```

```
INSERT INTO Course (CourseName, Credits)
VALUES ('Database Systems', 3), ('Computer Networks', 4);
```

```
INSERT INTO Enrollment (StudentID, CourseID, Grade)
VALUES (1, 1, 'A'), (2, 2, 'B');
```

3. Manipulating Data

Updating Records

```
UPDATE Student
SET CGPA = 3.85
WHERE StudentID = 1;
```

```
UPDATE Student
SET District = 'Sylhet'
WHERE Department = 'CSE';
```

Deleting Records

```
DELETE FROM Student
WHERE StudentID = 3;
```

```
DELETE FROM Enrollment
WHERE Grade = 'B';
```

Querying Data (SELECT)

```
-- Retrieve all students
SELECT * FROM Student;
```

```
-- Retrieve students with CGPA above 3.5
SELECT Name, CGPA FROM Student WHERE CGPA > 3.5;
```

```
-- Distinct Districts
SELECT DISTINCT District FROM Student;
```

```
-- Sorting by CGPA
SELECT Name, Department, CGPA FROM Student ORDER BY CGPA DESC;
```

4. Advanced Manipulation Examples

```
-- UNION Example
SELECT Name, Department FROM Student WHERE Department = 'CSE'
UNION
SELECT Name, Department FROM Student WHERE Department = 'EEE';

-- UNION ALL Example
SELECT Name, Department FROM Student WHERE Department = 'CSE'
UNION ALL
SELECT Name, Department FROM Student WHERE Department = 'EEE';

-- CASE Example: Assigning Grades
SELECT Name, CGPA,
       CASE
         WHEN CGPA >= 3.75 THEN 'A'
         WHEN CGPA >= 3.50 THEN 'B+'
         WHEN CGPA >= 3.00 THEN 'B'
         ELSE 'C'
       END AS Grade
FROM Student;

-- Aggregate Functions
SELECT Department, AVG(CGPA) AS AvgCGPA, COUNT(StudentID) AS TotalStudents
FROM Student
GROUP BY Department;
```

5. Conclusion

This document demonstrates the implementation and manipulation of a relational database in SQL Server using the Student database. The ER diagram shows the relationship between students, courses, and enrollments. Through SQL commands, we created tables, inserted records, updated and deleted data, and performed queries using SELECT, DISTINCT, UNION, UNION ALL, CASE, and aggregate functions. These manipulations highlight the flexibility and power of relational databases in managing structured data effectively.